LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE-6

Jenskin

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PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

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PRAKTIKUM

Jenkins Installation (WSL)

1. Masuk ke wsl dan pindah ke cd home

```
C:\Users\souya>wsl
souya@SouyaDream:/mnt/c/Users/souya$ cd ~
souya@SouyaDream:~$ sudo su
[sudo] password for souya:
root@SouyaDream:/home/souya#
```

2. Install JDK Java

```
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jimage to provide /usr/bin/jimage (jimage) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jinfo to provide /usr/bin/jinfo (jinfo) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jlink to provide /usr/bin/jlink (jlink) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmap to provide /usr/bin/jmap (jmap) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jmod to provide /usr/bin/jmod (jmod) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jps to provide /usr/bin/jps (jps) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jrunscript to provide /usr/bin/jrunscript (jrunscript) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jshell to provide /usr/bin/jshell (jshell) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstack to provide /usr/bin/jstack (jstack) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstat to provide /usr/bin/jstat (jstat) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jstatd to provide /usr/bin/jstatd (jstatd) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/serialver to provide /usr/bin/serialver (serialver) in auto mode
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-17-jdk:amd64 (17.0.18+8-1~24.04.1) ...
update-alternatives: using /usr/lib/jvm/java-17-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
root@SouyaDream:/home/souya#
```

3. Cek versi java

```
root@SouyaDream:/home/souya# java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
root@SouyaDream:/home/souya#
```

4. Menambahkan Jenskins Repository Key

```
root@SouyaDream:/home/souya# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-03-12 20:10:45--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.46.133, 2a04:4e42:7b::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.46.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring.asc          100%[===================================================================================>]   1.64K  --.-KB/s    in 0s

2026-03-12 20:10:46 (15.4 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]
```

5. Tambahkan Jenkins Repository

```
root@SouyaDream:/home/souya# echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

6. Install Jenkins

```
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 95.8 MB/96.0 MB of archives.
After this operation, 97.2 MB of additional disk space will be used.
Get:1 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.541.2 [95.8 MB]
Fetched 95.8 MB in 19s (4917 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 44025 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.541.2_all.deb ...
Unpacking jenkins (2.541.2) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Setting up jenkins (2.541.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
Processing triggers for man-db (2.12.0-4build2) ...
```

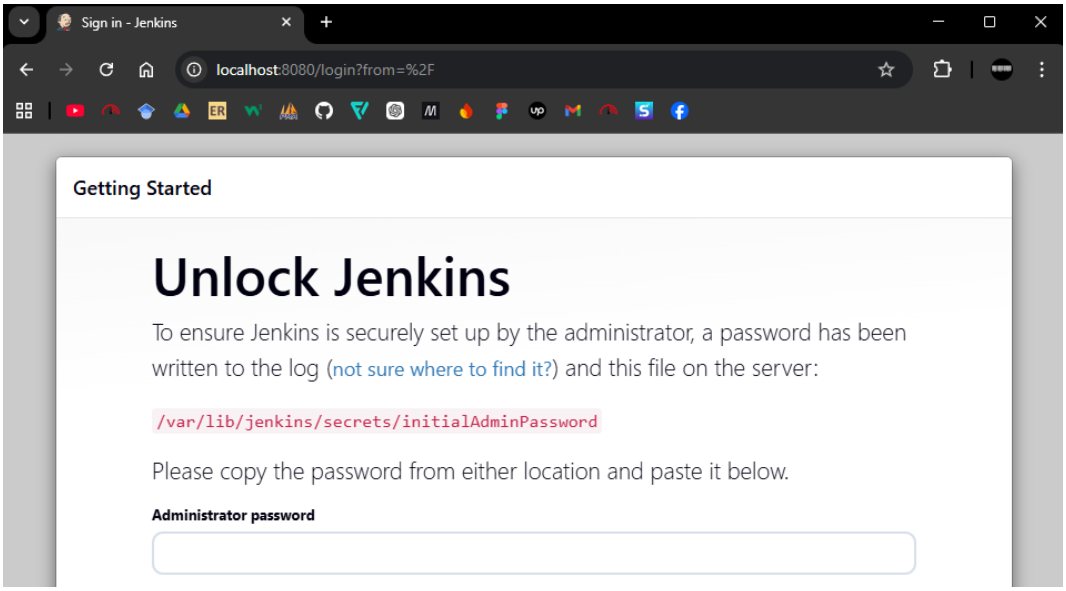
7. Jalankan Jenkins

```
root@SouyaDream:/home/souya# sudo systemctl start jenkins
root@SouyaDream:/home/souya# ls
docker_compose  laravel-docker  mysql-data  php
root@SouyaDream:/home/souya# sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@SouyaDream:/home/souya# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-12 20:19:53 WIB; 1min 13s ago
```

8. Cek apakah Jenskins bekerja

```
root@SouyaDream:/home/souya# sudo lsof -i :8080
COMMAND   PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
java     3253 jenkins    8u  IPv6  19338      0t0  TCP *:http-alt (LISTEN)
```

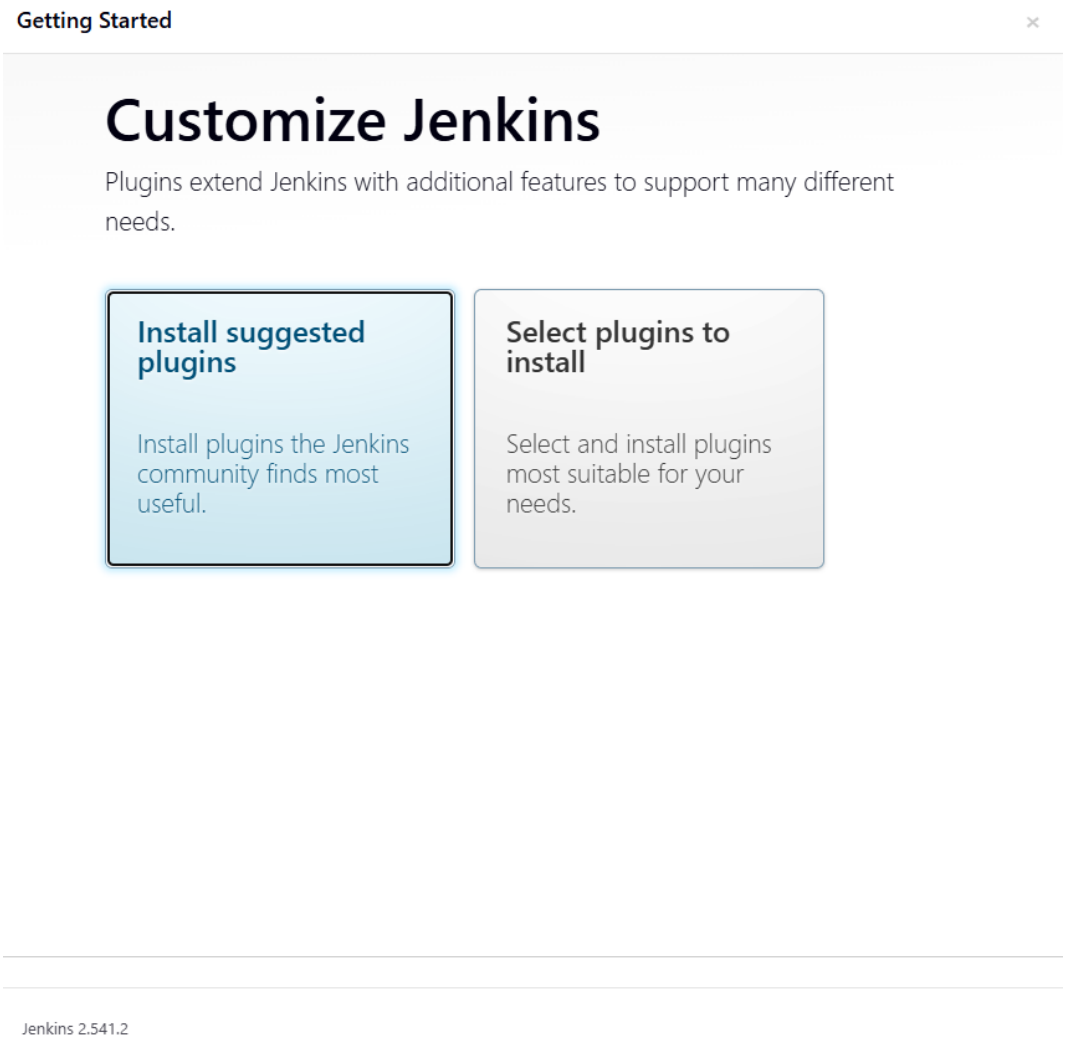
9. Dashboard Jenskins



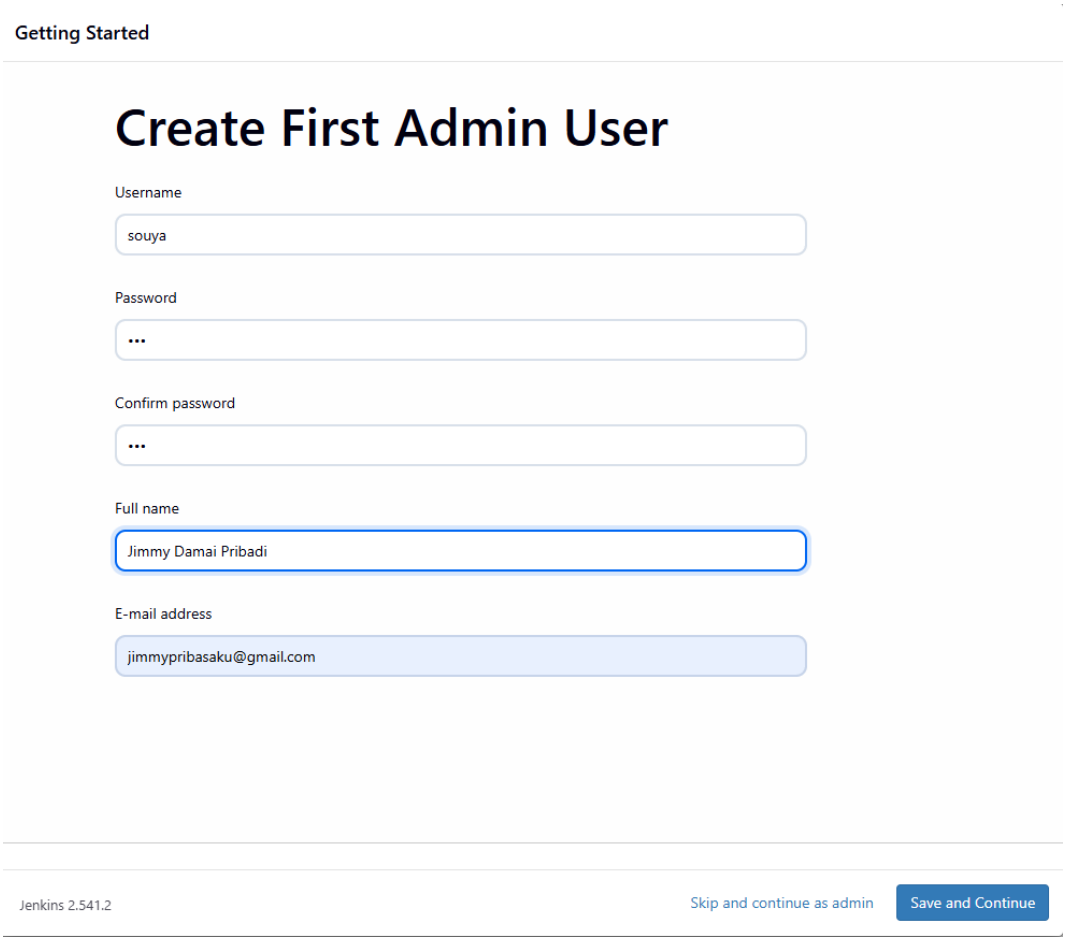
10. Ambil password awal Jenkins

```
root@SouyaDream:/home/souya# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
76cb4f849b644517835641027c4fded7
```

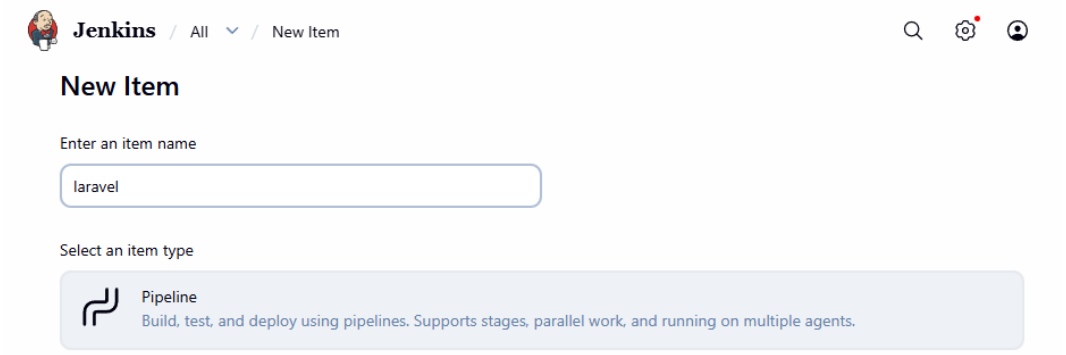
11. Pilih Install Suggested Plugins



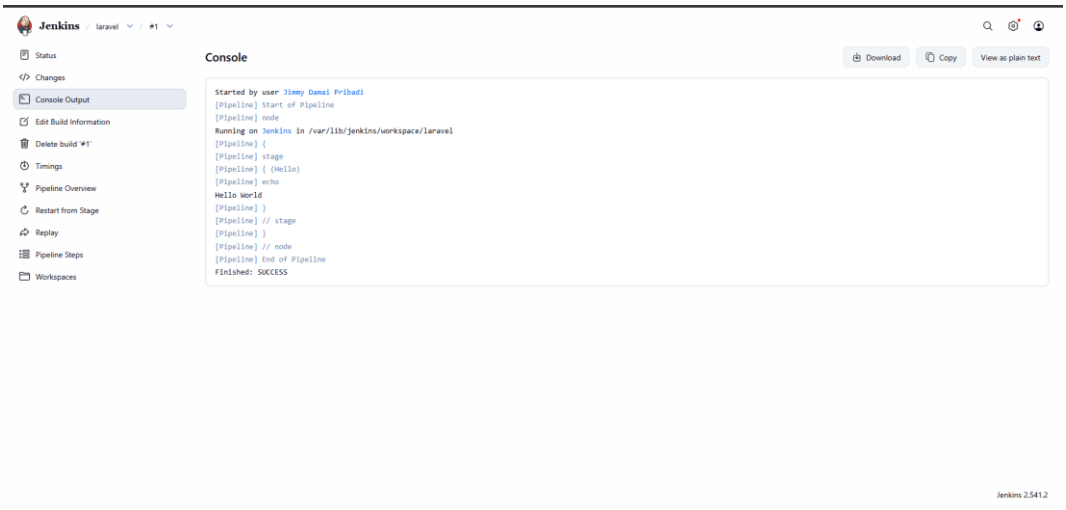
12. Register akun



13. Create Pipeline



14. Setelah berhasil di Build maka akan muncul seperti ini



Laravel Project Setup

1. Buat Folder Web

```
souya@SouyaDream:/mnt/c/Users/souya/Documents$ mkdir website
souya@SouyaDream:/mnt/c/Users/souya/Documents$ cd website
souya@SouyaDream:/mnt/c/Users/souya/Documents/website$ mkdir laravel_jenkins
```

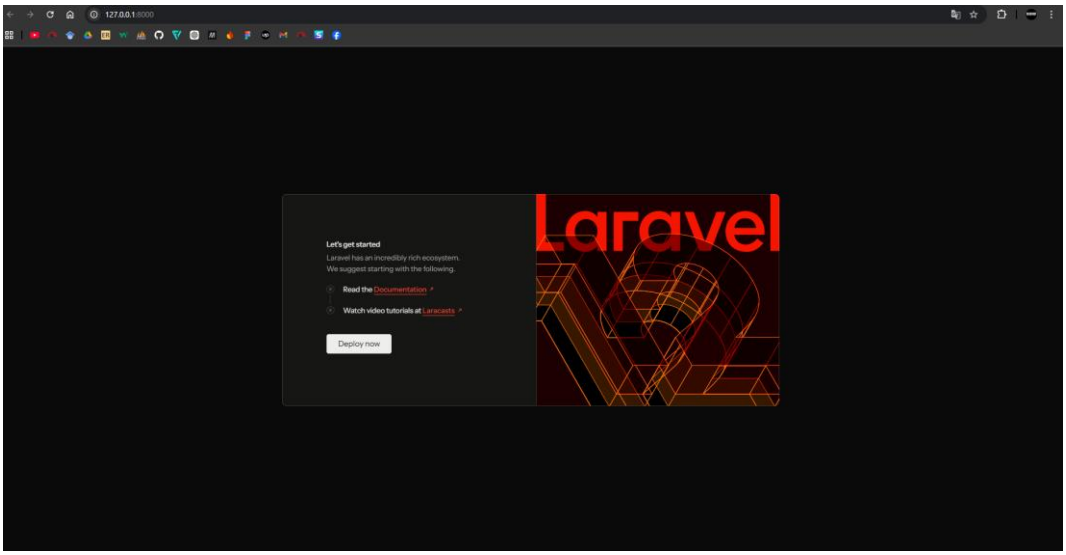
2. Install Laravel

```
souya@SouyaDream:/mnt/c/Users/souya/Documents/website$ composer create-project laravel/laravel laravel-ci
Creating a "laravel/laravel" project at "./laravel-ci"
Installing laravel/laravel (v12.12.1)
  - Downloading laravel/laravel (v12.12.1)
  - Installing laravel/laravel (v12.12.1): Extracting archive
```

3. Setup Environment

```
souya@SouyaDream:/mnt/c/Users/souya/Documents/website$ cd laravel-ci
souya@SouyaDream:/mnt/c/Users/souya/Documents/website/laravel-ci$ copy .env.example .env
php artisan key:generate
Command 'copy' not found, did you mean:
  command 'copay' from snap copay (12.5.6)
  command 'cozy' from snap cozy (1.3.0)
  command 'opy' from snap opy (latest)
  command 'rcopy' from deb rdmacm-utils (50.0-2ubuntu0.2)
  command 'hcopy' from deb hfsutils (3.2.6-15build2)
  command 'copyq' from deb copyq (7.1.0-1)
  command 'ropy' from deb libdisorder-tools (0.0.2+git20130809.8062ee1-4)
  command 'bcopy' from deb bacula-sd (9.6.7-7)
  command 'fcopy' from deb fai-client (6.0.5ubuntu1)
  command 'mcopy' from deb mtools (4.0.43-1)
See 'snap info <snapname>' for additional versions.

   INFO  Application key set successfully.
```

4. Laravel berhasil terinstall



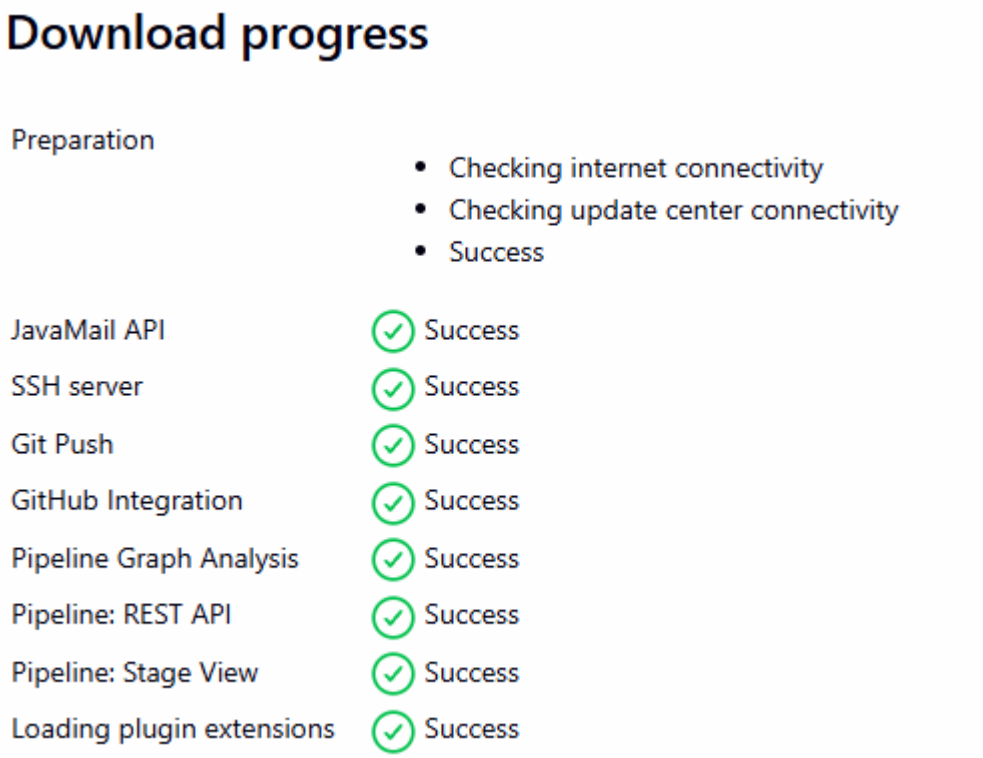
CI/CD using GitHub Actions

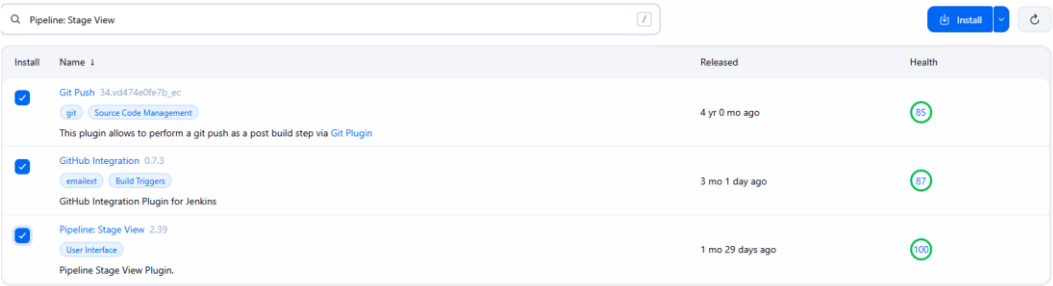
1. Isi file yml dengan konfigurasi ini

```
  GNU nano 7.2
name: Laravel CI

on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]

jobs:
  laravel-tests:

    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Setup PHP
        uses: shivammathur/setup-php@v2
        with:
          php-version: 8.2

      - name: Install dependencies
        run: composer install --no-interaction --prefer-dist

      - name: Copy ENV
        run: cp .env.example .env

      - name: Generate key
        run: php artisan key:generate

      - name: Run tests
        run: php artisan test
```

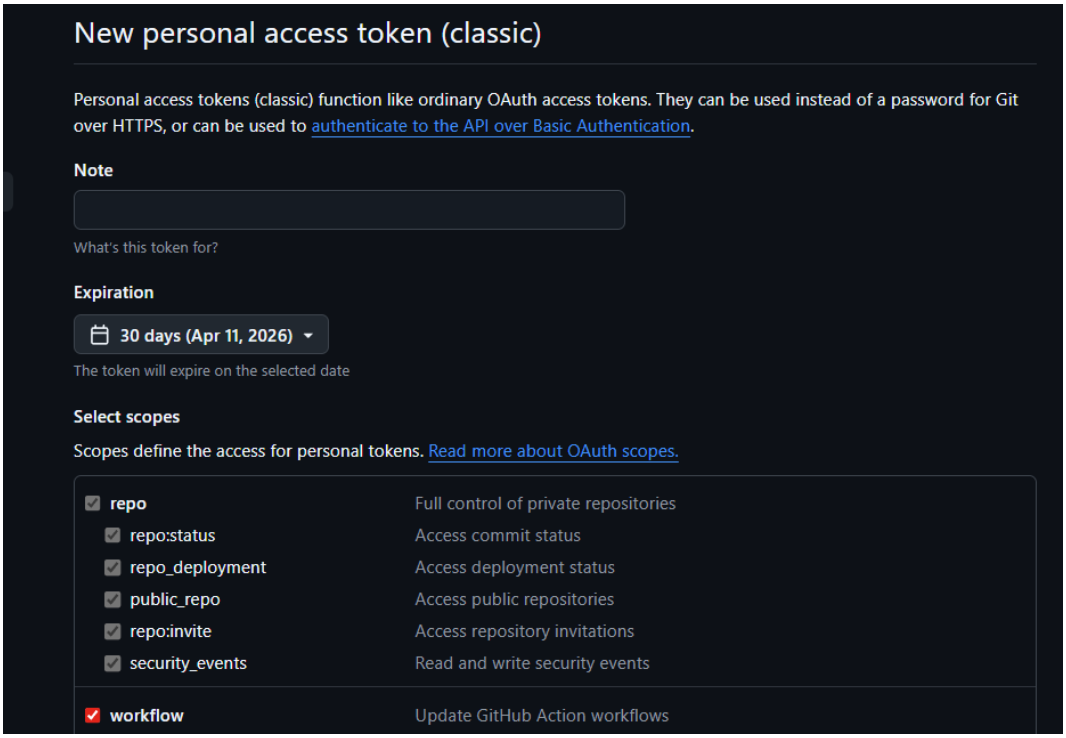
CI/CD using Jenkins

1. Menginstall Plugins

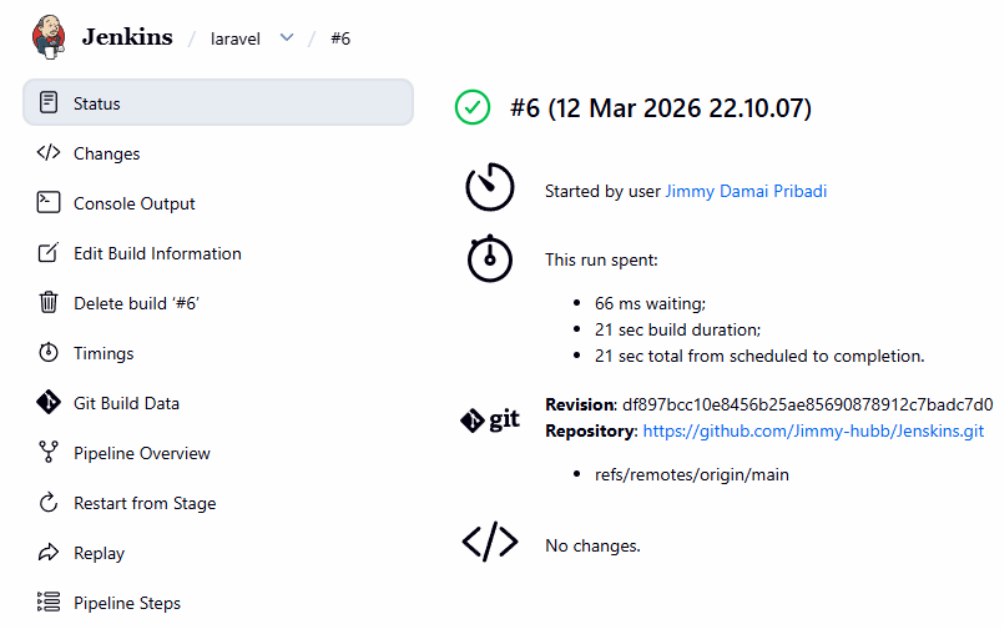




GitHub Credential Setup



Push di github

```
souya@SouyaDream:/mnt/c/Users/souya/Documents/website/laravel-ci$ git push -u origin main
Username for 'https://github.com': Jimmy-hubb
Password for 'https://Jimmy-hubb@github.com':
Enumerating objects: 80, done.
Counting objects: 100% (80/80), done.
Delta compression using up to 16 threads
Compressing objects: 100% (61/61), done.
Writing objects: 100% (80/80), 73.22 KiB | 191.00 KiB/s, done.
Total 80 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Jimmy-hubb/Jenskins.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Pipeline berhasil di build dari clone repository github saya